

Executive Summary

Ethiopian mobile banking is growing fast — and so is user frustration. This report presents preliminary findings from an ongoing analysis of 1,837 Google Play Store reviews across three major Ethiopian banks: Commercial Bank of Ethiopia (CBE), Bank of Abyssinia (BOA), and Dashen Bank. Using automated sentiment classification and thematic keyword extraction, we surface the clearest signals of what users love, what they complain about, and where each bank should focus its product investment.

Early findings show that CBE leads in user satisfaction (avg. 3.92 stars), while BOA faces the most negative sentiment (256 negative reviews vs. 212 positive). Across all three banks, app stability and transaction reliability emerge as the dominant pain points.

1. Data Collection Methodology

1.1 Scraping Approach

Reviews were collected using the google-play-scraper Python library, which interfaces with the Google Play Store’s internal API. We targeted the three most widely used mobile banking apps in Ethiopia:

Bank	App Package ID	Reviews Collected
Commercial Bank of Ethiopia	com.combanketh.mobilebanking	453
Bank of Abyssinia	com.boa.boaMobileBanking	499
Dashen Bank	com.cr2.amolelight + com.dashen.dashensuperapp	885

Total: 1,837 reviews (exceeding the 1,200 minimum requirement).

For Dashen Bank, two apps were scraped — the legacy Dashen Mobile app (formerly Amole Lite) and the newly launched Dashen Super App (January 2025) — and merged under a single label. Reviews were collected in English, filtered by country (Ethiopia), and sorted by newest first to capture the most recent user experience.

1.2 Preprocessing Steps

Raw data underwent the following cleaning steps:

- Deduplication: 468 duplicate reviews removed (same text + same bank)
- Missing value handling: Zero reviews dropped for missing text or rating
- Date normalization: All dates converted to YYYY-MM-DD format
- Column standardization: Five required columns retained

1.3 Data Quality Summary

Metric	Value
Total reviews collected	1,837
Duplicate reviews removed	468
Reviews with missing text	0
Reviews with missing rating	0
Missing data rate	less than 1%
Date range	2022 - 2026

1.4 Limitations

- English-only reviews: Many Ethiopian users write in Amharic. These were excluded, which may introduce sampling bias toward more tech-savvy, English-speaking users.
- Recency bias: Sorting by newest first means older reviews are underrepresented.
- Platform limitation: Only Google Play reviews were collected.

2. Early Sentiment Findings

2.1 Methodology

Sentiment classification was performed using distilbert-base-uncased-finetuned-sst-2-english, a transformer model fine-tuned on the Stanford Sentiment Treebank (SST-2). Each review was classified as POSITIVE or NEGATIVE with a confidence score (0-1). Reviews where confidence was below 70% were labeled NEUTRAL.

DistilBERT was chosen over lexicon-based tools (VADER, TextBlob) because it understands context. For example, it correctly classifies “the app would be great if it did not crash every time” as NEGATIVE, which rule-based tools often miss.

2.2 Overall Sentiment Distribution

Bank	Positive	Neutral	Negative	Total
Commercial Bank of Ethiopia	259 (57%)	21 (5%)	173 (38%)	453
Bank of Abyssinia	212 (42%)	31 (6%)	256 (51%)	499
Dashen Bank	506 (57%)	39 (4%)	340 (38%)	885

Key insight: BOA is the only bank where negative reviews outnumber positive ones.

2.3 Average Star Ratings

Bank	Average Rating
Commercial Bank of Ethiopia	3.92
Dashen Bank	3.86
Bank of Abyssinia	3.28

Bank	Average Rating
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BOA's 3.28 average is consistent with its negative sentiment majority. These two independent signals reinforce each other and point to a genuine product quality gap.

3. Early Thematic Findings

Reviews were assigned to one of five business-relevant themes using keyword matching:

Theme	Description
Account Access Issues	Login, OTP, password, biometric problems
Transaction Performance	Transfer speed, payment failures, delays
App Stability and Performance	Crashes, freezes, failed updates
UI and User Experience	Interface design, ease of use, navigation
Customer Support	Response quality, branch support, complaints

3.1 Preliminary Theme Insights

Early keyword analysis reveals distinct complaint profiles per bank:

- CBE negatives cluster around: transaction, transfer, account — payment reliability is the primary pain point
- BOA negatives cluster around: update, mobile, time — app updates appear to be breaking functionality
- Dashen negatives cluster around: slow, update, need — performance and missing features are the key concerns

4. Blockers and Plan for Final Submission

4.1 Blockers Encountered

Blocker	Resolution
Dashen Bank original app ID returned 0 reviews	Identified two active Dashen apps; scraped both and merged
TF-IDF keywords included noise (phone models, Amharic)	Added alphabetic-only token filter and frequency thresholds
Database tests failed in CI environment	Tests use <code>pytest.skip()</code> gracefully when DB is unavailable

4.2 Plan for Final Submission

Task	Status	Remaining Work
Task 1: Scraping and Preprocessing	Complete	None
Task 2: Sentiment and Thematic Analysis	Complete	Refine theme grouping
Task 3: PostgreSQL Database	Complete	Fix CI database credentials
Task 4: Visualizations and Report	In Progress	Word clouds, recommendations
Unit Tests	19 of 24 passing	Fix DB connection for remaining 5

5. Conclusion

This interim analysis demonstrates that automated review mining can surface actionable product intelligence at scale. With 1,837 reviews processed, sentiment classified, and themes assigned, the pipeline is production-ready. The final report will translate these findings into concrete, prioritized recommendations for CBE, BOA, and Dashen Bank product teams.

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